

ITAI1371 Regression Final Project Report

Introduction

This project uses machine learning to predict a number that shows how much a person's diabetes got worse after one year. The dataset used in this project is the diabetes dataset, which contains 442 rows of data, 10 input features, and 1 target column called `target`.

The goal of this project was to train several regression models, compare how well they performed, and then test two ensemble methods to see whether combining models would improve the predictions.

Data and Preprocessing

The dataset contains 10 numeric features and 1 target value. The features are age, sex, body mass index (BMI), average blood pressure (BP), and six blood measurement features labeled `s1` through `s6`.

Because the dataset is already clean and numeric, no major preprocessing such as filling in missing values or changing text categories into numbers was needed. The main preprocessing step was preparing the features and target and then splitting the data into training, validation, and test sets.

The data was split into three parts: 70 percent for training, 15 percent for validation, and 15 percent for testing. The training set was used to teach the models, the validation set was used to compare models and choose the best ones, and the test set was used for the final check on new data.

Models Used

Several regression models were trained in this project. Each model tries to predict the target value in a different way.

- **Linear Regression** is a simple model that tries to find the best straight-line relationship between the input features and the target.
- **Decision Tree Regressor** is a model that makes predictions by splitting the data into branches based on feature values.
- **Random Forest Regressor** combines many decision trees and averages their answers.
- **Gradient Boosting Regressor** builds many small trees in sequence and keeps improving its mistakes.
- **K-Nearest Neighbors Regressor (KNN)** predicts a value by looking at the most similar nearby data points.

- **Extreme Gradient Boosting (XGBoost)** is an advanced boosting method that improves prediction step by step.
- **Voting_Top3** is an ensemble model that averages the predictions of the three best base models.
- **Bayesian Ensemble** uses the predictions of the top three models as inputs to a Bayesian Ridge model.

Evaluation Metrics

The models were compared using **Mean Absolute Error (MAE)**, **Mean Squared Error (MSE)**, and **R-squared (R²)** on both the validation and test sets.

- **Mean Absolute Error (MAE)** shows the average size of the model's mistakes. Smaller MAE means better predictions.
- **Mean Squared Error (MSE)** also measures mistakes, but it gives more punishment to big mistakes. Smaller MSE is better.
- **R-squared (R²)** shows how well the model explains the pattern in the data. Higher R² means a better fit.

Results

The table below summarizes the results for all models.

Model	Val MAE	Val MSE	Val R ²	Test MAE	Test MSE	Test R ²
LinearRegression	38.22	2499.32	0.511	45.57	3139.37	0.447
DecisionTree	61.56	6207.92	-0.214	58.49	5195.27	0.085
RandomForest	39.25	2704.61	0.471	45.22	2904.35	0.488
GradientBoosting	42.35	2858.23	0.441	47.07	3281.49	0.422
KNN	41.46	2969.18	0.419	46.51	3471.28	0.389
XGBoost	41.84	3018.40	0.410	45.40	3318.62	0.416
Voting_Top3	38.82	2532.14	0.505	45.06	2953.74	0.480
Bayesian_Ensemble	42.46	3169.63	0.380	45.54	3069.84	0.459

On the validation set, **Linear Regression** performed best. It had the lowest validation Mean Squared Error (MSE) of 2499.32 and the highest validation R-squared (R^2) of 0.511. This means it was the strongest model during the model-selection stage.

On the test set, **Random Forest Regressor** performed best overall. It had the lowest test Mean Squared Error (MSE) of 2904.35 and the highest test R-squared (R^2) of 0.488. This means it gave the best predictions on new, unseen data.

Ensemble Model Analysis

Two ensemble methods were tested in this project.

The first ensemble was **Voting_Top3**, which averaged the predictions of the three best base models. Its test Mean Squared Error (MSE) was 2953.74 and its test R-squared (R^2) was 0.480, which made it one of the strongest models in the project. Even though it did not beat Random Forest Regressor, it still performed better than many individual models such as Gradient Boosting Regressor, K-Nearest Neighbors Regressor (KNN), and Extreme Gradient Boosting (XGBoost).

The second ensemble was the **Bayesian Ensemble**. Its test Mean Squared Error (MSE) was 3069.84 and its test R-squared (R^2) was 0.459. This model performed reasonably well, but it did not beat the simpler Voting_Top3 method.

Which Model Performed Best and Why

The best model in this project was **Random Forest Regressor**, because it achieved the highest test R-squared (R^2) and the lowest test Mean Squared Error (MSE). This is important because the test set is the best measure of how well a model works on new data.

A likely reason Random Forest Regressor performed best is that it can learn more complex patterns than Linear Regression. Linear Regression works well when the relationship between features and the target is mostly straight and simple, while Random Forest Regressor can capture nonlinear patterns by combining many decision trees. That extra flexibility likely helped it generalize better on the test set.

The results also show that simple ensembling can be useful. Voting_Top3 came very close to Random Forest Regressor, showing that averaging strong models can create stable predictions. However, the Bayesian Ensemble did not improve enough to justify its extra complexity in this case.

Overall Approach

The project followed a clear machine learning workflow. First, the data was prepared and split into training, validation, and testing sets. Next, multiple regression models were trained and compared

using Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2). After that, the top-performing models were combined into ensemble methods and evaluated again.

Visualizations such as the target distribution, correlation heatmap, model comparison bar charts, and predicted-versus-actual scatter plot were added to the notebook to make the results easier to understand. These visuals helped explain both the data and the model performance in a clearer way.

Limitations and Future Improvements

This project produced useful results, but there are still ways to improve it. The models were tested using one train/validation/test split, so the results could change slightly with a different split. Future work could use cross-validation, hyperparameter tuning, and feature engineering to improve model performance further.

Another possible improvement would be to try more advanced ensemble methods or compare the diabetes dataset with another regression dataset. This would help test whether the same models remain strong across different types of regression problems.

Simple Glossary

- **MAE:** Mean Absolute Error
- **MSE:** Mean Squared Error
- **R^2 :** R-squared
- **KNN:** K-Nearest Neighbors
- **XGBoost:** Extreme Gradient Boosting
- **Ensemble:** a method that combines more than one model
- **Validation set:** data used to compare models during development
- **Test set:** data used for the final evaluation

Conclusion

This regression project successfully trained and compared several machine learning models to predict diabetes progression. Linear Regression performed best on the validation set, but Random Forest Regressor performed best on the final test set, making it the strongest overall model for this dataset.

The Voting_Top3 ensemble also gave strong results and came close to Random Forest Regressor, while the Bayesian Ensemble did not improve over the simpler voting approach. Overall, the project

met the required steps of data preprocessing, splitting, training multiple models, validating them, building ensembles, and analyzing the final results in a way that is easy to understand.